

Functional Annotation of *nadC* (PP_0787, UniProt Q88PR1) in *Pseudomonas putida* KT2440

Gene/Protein Identity — Verified

- **UniProt:** Q88PR1 — "Probable nicotinate-nucleotide pyrophosphorylase [carboxylating]", EC 2.4.2.19; AltName **Quinolinate phosphoribosyltransferase [decarboxylating]**.
- **Gene / locus:** *nadC* / PP_0787.
- **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125).
- **Family / domains:** NadC/ModD family; Aldolase_TIM (IPR013785), NadC (IPR004393), NadC/ModD (IPR027277), Nicotinate_pribotase-like_C (IPR036068), Quinolinate_PRibosylTase_N (IPR037128).

Verification: The gene symbol *nadC*, the enzyme name (quinolinate phosphoribosyltransferase / nicotinate-nucleotide pyrophosphorylase [carboxylating]), the EC number 2.4.2.19, and the domain architecture are fully mutually consistent and correspond to a single, deeply conserved enzyme of NAD⁺ biosynthesis. **This is an unambiguous identification.** The one caveat is that no biochemical or genetic study has been published on the *P. putida* KT2440 protein itself; its function is assigned by strong orthology to biochemically and structurally characterized NadC enzymes from *Salmonella*, *E. coli*, *Helicobacter pylori*, *Mycobacterium tuberculosis*, and eukaryotes. Given >30% sequence identity across this family and full conservation of catalytic residues and fold, the transfer of annotation is robust.

1. Summary (Answer to the Research Question)

nadC encodes **quinolinate phosphoribosyltransferase (QPRTase / QAPRTase; also called nicotinate-nucleotide pyrophosphorylase [carboxylating])**, EC 2.4.2.19. It catalyzes the committed, magnesium-dependent step of *de novo* NAD⁺ biosynthesis: the transfer of a phosphoribosyl group from 5-phosphoribosyl-1-pyrophosphate (PRPP) onto **quinolinic acid**,

coupled to decarboxylation, yielding **nicotinic acid mononucleotide (NaMN)**, CO_2 , and **pyrophosphate (PPi)**. It is a soluble **cytoplasmic** enzyme, a structural prototype of the **type II PRTase** family with a TIM-barrel catalytic fold, and functions as a dimer (active site at the subunit interface), frequently assembling into a hexameric trimer-of-dimers. It is the single, essential gateway converting the aspartate/tryptophan-derived intermediate quinolinate into the NAD^+ pathway.

2. Primary Function — The Catalyzed Reaction

Reaction:

quinolinic acid (pyridine-2,3-dicarboxylic acid) + PRPP + H^+ → nicotinic acid mononucleotide (NaMN) + CO_2 + PPi

The chemistry proceeds as a **phosphoribosyl transfer followed by decarboxylation** — the pyridine N1 of quinolinate is ribosylated by PRPP (displacing PPi), and the C3 carboxyl of the resulting intermediate is lost as CO_2 , producing NaMN [PMID 24949952]. The full stoichiometry (NaMN + CO_2 + PPi from PRPP + quinolinate) was established for the *Salmonella typhimurium* enzyme [PMID 11876660]. The reaction is strictly **divalent-metal dependent, with Mg^{2+} optimal** [PMID 24949952].

Substrate specificity. The substrate is the **dicarboxylic** pyridine quinolinate. Specificity is enforced by a cluster of cationic active-site residues (in *S. typhimurium*: Arg118, Arg152, Arg175, Lys185, His188) that recognize the 2,3-dicarboxylate. **Arg175 is catalytically essential** (R175A is inactive), while R152A and K185A shift activity toward the normally inactive monocarboxylate analogue nicotinic acid — K185A shows ~300-fold higher $k_{\text{cat}}/K_{\text{m}}$ for nicotinate than for quinolinate — demonstrating that these residues both catalyze the reaction and select quinolinate over related pyridines [PMID 20047306]. Consistent kinetic parameters have been measured for orthologs (e.g., *M. tuberculosis*: $K_{\text{m}}(\text{QA}) \approx 0.08$ mM, $K_{\text{m}}(\text{PRPP}) \approx 0.39$ mM) [PMID 24949952]; the QA analogue **pyrazinamide** inhibits the mycobacterial enzyme [PMID 24949952], and **phthalic acid** is a classic competitive inhibitor/QA mimic used in structural studies [PMID 24038671].

Kinetic mechanism. The *Salmonella* enzyme follows a **predominantly ordered mechanism**: quinolinate binds productively first (the E·QA complex is catalytically competent, E·PRPP is not), and **on-enzyme phosphoribosyl-transfer chemistry is rate-determining** [PMID 11876660].

3. Pathway Context

nadC is the third enzyme of the bacterial **aspartate → NAD⁺ *de novo* pathway**:

1. **NadB** (L-aspartate oxidase) oxidizes L-aspartate to iminoaspartate.
2. **NadA** (quinolinate synthase) condenses iminoaspartate with dihydroxyacetone phosphate to form **quinolinate**.
3. **NadC (this gene)** converts quinolinate + PRPP → **NaMN** + CO₂ + PPi.
4. **NadD** (NaMN adenylyltransferase) → nicotinic acid adenine dinucleotide (NaAD).
5. **NadE** (NAD synthetase) → **NAD⁺**.

Thus NadC sits at the junction where the ring precursor (quinolinate) enters the mononucleotide pool that all downstream NAD⁺ synthesis flows through. The *nadA/nadB/nadC* gene set is broadly conserved across bacteria; the products converge on the same NaMN node used by the salvage (Preiss–Handler) route [PMID 203571;].

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Regulation. Expression of the *de novo* genes (including *nadC*) is repressed by nicotinate through NrtR/NdnR-family transcriptional regulators (shown in *Corynebacterium glutamicum*: "NdnR acts as a transcriptional repressor of NAD *de novo* biosynthesis genes"), so cells down-regulate costly *de novo* synthesis when the salvage precursor nicotinate is available [PMID 20601509]. In *Bordetella*, an orphan *nadC* is controlled by a NadQ-family repressor and by nicotinic acid [PMID 27783449].

Confirmed in *P. putida* KT2440 (KEGG). The KT2440 genome encodes the **complete** *de novo* pathway, so NadC operates as a genuine biosynthetic enzyme (not an orphan salvage gene): *nadB*/L-aspartate oxidase = **PP_1426** (K00278), *nadA*/quinolinate synthase = **PP_1231** (K03517), **nadC** = **PP_0787** (K00767, EC 2.4.2.19, reaction R03348), *nadD*/NaMN adenylyltransferase = **PP_4810** (K00969), *nadE*/NAD synthetase = **PP_4869** (K01916). PP_0787 is assigned to KEGG module **M00115** ("NAD biosynthesis, aspartate ⇒ quinolinate ⇒ NAD") and pathways ppu00760

and ppu01240. It sits at genome position complement(902650..903498), flanked by *trxB* (PP_0786) and PP_0788 (unknown), and is **monocistronic** — the five *nad* genes are scattered rather than in a single operon.

Note specific to *P. putida* KT2440: this organism additionally possesses a well-known **nicotinate catabolic (nic) locus** that degrades nicotinic acid (via 6-hydroxynicotinate); the *P. putida* *nic* system is the reference for related organisms and modulates intracellular pyridine pools [PMID 29485696]. This is a distinct catabolic pathway; *nadC*/NadC belongs to the **anabolic** NAD⁺-biosynthetic route.

4. Physiological Importance (Essentiality)

NadC catalyzes "an indispensable step in NAD biosynthesis, one that is essential for cell survival in prokaryotes" [PMID 17763926]. Genetic evidence across species: - *S. typhimurium* *nadC* mutants are **nicotinate auxotrophs** that **accumulate and secrete quinolinate** (2–8 μmol/100 ml) and cannot grow on quinolinate [PMID 203571]. - *C. glutamicum* Δ *nadC* (and Δ *nadA*, Δ *nadS*) require exogenous nicotinate [PMID 20601509]. - Even in organisms that have lost *nadA*/*nadB*, an "orphan" *nadC* is retained to allow **quinolinate salvage**; quinolinate as sole NAD precursor requires *nadC* [PMID 27783449;

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Because it is essential and structurally distinct in bacteria vs. humans, bacterial NadC is pursued as an **antibacterial/anti-tuberculosis drug target** [PMID 17763926;

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5. Structure and Localization

- **Fold / class:** QPRTase is the defining structural prototype of the **type II PRTase** family ("the only type II PRTase whose X-ray structure is known") [PMID 11876660]. The catalytic domain is an (β/α)₈ **TIM-barrel** (Aldolase_TIM, IPR013785) with an N-terminal NadC domain and a nicotinate-phosphoribosyltransferase-like C-terminal region — exactly the domain set annotated for Q88PR1.

- **Quaternary structure:** The minimal functional unit is a **dimer**, with the **active site formed at the monomer–monomer interface** (e.g., *S. typhimurium*, *H. pylori*). Many QPRTases (human, porcine, yeast, *H. pylori*) assemble into a **hexamer (trimer of dimers)** that is required for full activity/stability; e.g., disrupting the *H. pylori* $\alpha 7$ – $\beta 8$ loop hexamerization interface (Phe181Pro) abolishes activity [PMID 17763926;

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- **Subcellular localization:** The enzyme is a **soluble cytoplasmic protein**. It carries no signal peptide or transmembrane region and is routinely purified from the cytosol; it carries out its function in the **cytoplasm** as part of central NAD⁺ metabolism.

5b. Direct Sequence Verification of Q88PR1 (protein-specific evidence)

Beyond generic family annotation, I aligned the *P. putida* KT2440 NadC sequence (Q88PR1, 282 aa) against the biochemically/structurally characterized *Salmonella typhimurium* QAPRTase (P30012, 297 aa) by global Needleman–Wunsch alignment:

- **61.2% amino-acid identity** (172/281 aligned positions) — clear orthology, well above the ~30% typical for QPRTase cross-family comparisons.
- **All catalytic signature motifs are conserved verbatim in Q88PR1:**
 - **DTRKTLPLGR** — substrate/PRPP-binding loop bearing the catalytic Arg and Lys (identical in both proteins).
 - **CGG** — phosphate/metal-coordinating loop.
 - **NHR** — conserved His–Arg motif.
 - **GADIIMLD** — divalent-metal/catalytic-aspartate loop.
 - **(A)ETGVD** — TIM-barrel signature.
 - **ISIGA** — C-terminal PRPP-binding glycine motif (ISVGA in *Salmonella*).
- **No signal peptide or transmembrane segment** is present, consistent with a soluble cytoplasmic enzyme.

This demonstrates directly — not merely by database inference — that Q88PR1 possesses an intact QPRTase active site and is a catalytically competent quinolinate phosphoribosyltransferase, strongly validating the functional annotation for the *P. putida* protein itself.

6. Supported vs. Refuted Hypotheses

Supported (by orthology + conserved biochemistry/structure/genetics): - H1: Q88PR1/NadC is quinolinate phosphoribosyltransferase (EC 2.4.2.19). ✓ - H2: It catalyzes quinolinate + PRPP → NaMN + CO₂ + PPi, Mg²⁺-dependent, ordered mechanism. ✓ - H3: It is the committed, essential step of *de novo* NAD⁺ biosynthesis (nadB→nadA→**nadC**→nadD→nadE). ✓ - H4: It is a cytoplasmic type II PRTase / TIM-barrel enzyme acting as a dimer (± hexamer). ✓ - H5: Substrate selectivity for the dicarboxylate quinolinate is enforced by conserved cationic active-site residues. ✓ - H6: The Q88PR1 sequence itself retains an intact QPRTase active site (61% identity to Salmonella NadC; all catalytic motifs conserved). ✓ (direct sequence evidence) - H7: In *P. putida* KT2440 specifically, nadC functions in a complete *de novo* NAD⁺ pathway (nadB PP_1426, nadA PP_1231, nadC PP_0787, nadD PP_4810, nadE PP_4869 all present; KEGG module M00115). ✓ (organism-specific genomic evidence)

Refuted / not supported: - The gene is NOT a nicotinate/NAD *catabolic* enzyme (the separate *P. putida* nic locus performs nicotinate degradation). ✗ - No evidence for a membrane-bound or secreted role. ✗

7. Limitations and Future Directions

- **No direct study of the *P. putida* KT2440 protein.** All biochemical, kinetic, structural, and genetic evidence is from orthologs; annotation transfer is well justified by conservation but not experimentally confirmed for PP_0787.
- Future work: recombinant expression and kinetic characterization of PP_0787; a *P. putida* Δ nadC growth/auxotrophy test with quinolinate; and an AlphaFold/experimental structure to confirm the TIM-barrel dimer and active-site cationic cluster in the *Pseudomonas* enzyme.

Key References

- P 11876660

 — Cao, Pietrak, Grubmeyer (2002): kinetic mechanism of *Salmonella* QAPRTase; type II PRTase; reaction stoichiometry.
- P 20047306

 — Bello & Grubmeyer (2010): cationic active-site residues and substrate specificity.
- P 24949952

 — Kim et al. (2014): *M. tuberculosis* QAPRTase biochemistry (Mg²⁺, kinetics, decarboxylation, pyrazinamide inhibition).
- P 203571

 — Foster & Moat (1978): *nadC* = QAPRTase; auxotrophy and quinolinate accumulation.
- P 17763926

 — Kim et al. (2007): *H. pylori* NadC hexamer (trimer of dimers), essentiality.
- P 26805589

 23626766 / 24038671 — human/porcine QPRT structures; hexameric mechanism; phthalate inhibitor.
- P 20601509

 — Teramoto et al. (2010): NdnR repression of *de novo* NAD genes.
- P 27783449

 — Brickman et al. (2017): orphan *nadC*, quinolinate salvage.
- P 23204464

 — Sorci et al. (2013): NadC function and NAD metabolism reconstruction.
- P 29485696

 — Brickman & Armstrong (2018): *P. putida* KT2440 nicotinate catabolism (distinct pathway).